

PHARMACOLOGICAL AND TOXICOLOGICAL SCREENING METHODS-II

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

10x2 = 20

- (a) Explain acute toxicity study
- (b) What is Dose progression factor?
- (c) What is HERG assay?
- (d) Suggest some important applications of toxicokinetic studies.
- (e) Define Ames test
- (f) Why do we carry out skin sensitization test?
- (g) Suggest inhalational studies according to regulatory guidelines
- (h) What is the importance of study of genotoxicity?
- (i) Suggest the major features of GLP
- (j) What are roles and responsibilities of EPA?

SECTION B

2. Attempt any *two* of the following:

2 x 10 = 20

- (a) Describe briefly oral and dermal toxicity studies. Explain significance of toxicity study for drugs undergoing saturation kinetics
- (b) Describe briefly Schedule Y for conducting toxicity studies. Explain female reproductive toxicity studies
- (c) Write species used, list of observations and report content of the following test as per OECD guidelines: i) Acute eye irritation ii) Skin sensitization

SECTION C

3. Attempt any *five* part of the following:

5 x 7 = 35

- (a) Enlist CNS assessment as per safety pharmacological studies and explain any one in detail.
- (b) Explain test procedure and clinical observations of prenatal developmental toxicity study
- (c) Suggest the various alternative methods of animal toxicity study
- (d) Justify the various studies involved in IND submission
- (e) Explain in detail invivo carcinogenicity study for newer molecules
- (f) Explain in detail OECD guideline 407 and ICH guidelines.
- (g) Write short notes on "Alternative methods to animal toxicity testing."